

LESSON 7-3 • SESSION 2 • PRACTICE • FORM B

Absolute Value as Total Distance

Learning target – I can use absolute value to find the total distance between a negative and a positive reading. • 6.NS.C.7c

REMEMBER

- **Opposite sides of 0:** add the absolute values. From -8 to 3 : $8 + 3 = 11$.
- **Same side of 0:** subtract the smaller absolute value from the larger. From -12 to -5 : $12 - 5 = 7$.
- A distance is **never negative**. Sketch a number line to see where 0 is.

WORKED EXAMPLEHow far is it from -8 to 3 on a number line?**Step 1** 0 is between -8 and 3 , so they are on opposite sides.**Step 2** $|-8| = 8$ and $|3| = 3$.**Step 3** Add the distances: $8 + 3 = 11$.**Answer 11 units****WORDS** absolute value (*valor absoluto*) • distance (*distancia*) • opposite sides (*lados opuestos*)**PRACTICE**

show your work • label units

1. Find the distance between each pair of numbers.

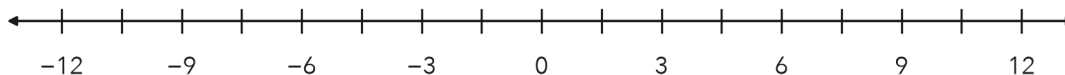
a. -18 and 5 _____

b. 7 and 25 _____

c. -20 and -6 _____

d. -8 and 3 _____

- 2.
- Freezer Thermometer.**
- A freezer read
- -10.5
- degrees before defrosting and
- 7.5
- degrees after. Zero is the freezing point of water.

Hint Zero is between the readings, so add.

- a. Plot both readings. Absolute values: Before ____ After ____

- b. Find the total change. _____ degrees

- 3.
- Garage Elevator.**
- An elevator goes from floor
- -6
- up to floor
- 8
- . How many floors does it travel?

Hint Count 6 floors up to 0, then 8 more.

Show your work.

Answer _____ floors

- 4.
- Drone Climb.**
- A drone climbs from
- 8
- m above the ground to
- 23
- m above the ground. How far did it climb?

Hint Both heights are above 0, so subtract.

Show your work.

Answer _____ m

- 5.
- Peak and Trench.**
- A peak rises
- 1.8
- km above the sea surface. A trench floor lies
- 4.7
- km below the sea surface. Find the vertical distance from the trench floor to the peak.

Answer _____ km

6. **Pump Move.** A well pump moved from -12 m to -47 m. Find the distance it moved. Why do you not add the absolute values?

Hint Both readings are below 0.

Show your work.

Answer _____ m

7. **Club Balance.** A club account moved from -48 dollars in May to 27 dollars in June. Find the total change.

Show your work.

Answer _____ dollars

8. **Mountain and Valley.** Trail 1 starts at -150 m in a valley and climbs to a peak at 280 m. Trail 2 runs from 60 m up to 280 m.

a. Total climb of Trail 1: _____ m

b. Total climb of Trail 2: _____ m

c. Why is Trail 1 longer if both end at 280 m? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Malik's Work.** A lift moves from floor -7 to floor 10. Malik wrote: $|10| - |-7| = 10 - 7 = 3$ floors. Explain his mistake. Then find the correct distance.

TEST PRACTICE

choose the best answer

10. The distance from 8 to 23 is –

A 31

B 15

C -15

D 23

11. The distance from -6 to 4 is –

A 2

B -10

C 10

D 24

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

75 U	15 S	18 A	6.5 O	14 B	220 E	35 L	21 Z	430 T	59 Q
#2b	#3	#4	#5	#6	#7	#8a	#8b		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about # _____